



POLITECHNIKA
LUBELSKA
WYDZIAŁ ELEKTROTECHNIKI
I INFORMATYKI

Temat projektu:

Aplikacja analizująca wpływ wiadomości na zwroty kryptowalut

Skład Grupy

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Opis projektu

Projekt analizuje statystyczny wpływ wiadomości na dzienne zwroty kryptowalut. Dane pobierane są z Yahoo Finance, a dane o artykułach z lokalnego pliku JSON

Użyte technologie

Backend

- Python 3.10(z rozszerzeniami wypisanymi w pliku requirements.txt)

- REST

- SQLite

Autentykacja

- JWT

- bcrypt

Dane

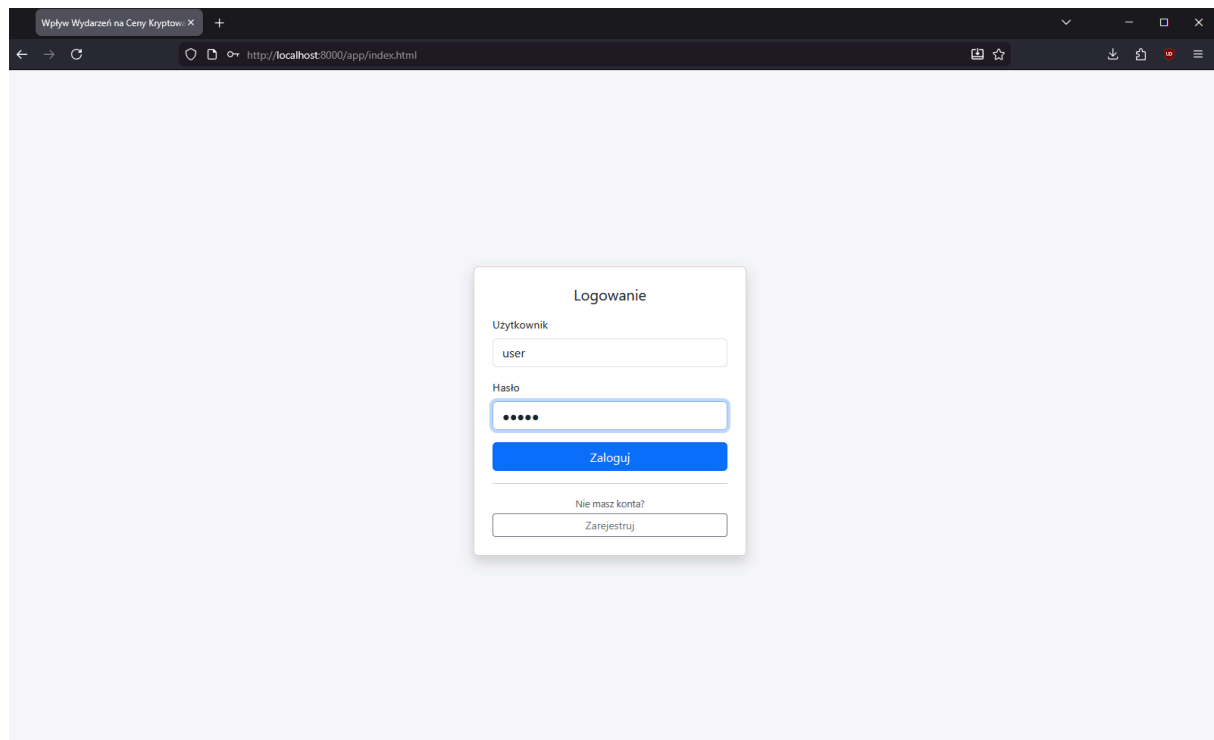
- JSON,XML,YAML

Frontend

- HTML/CSS/JavaScript

Frontend/UI

Panel logowania/rejestracji



Panel główny

Wyływ Wydarzeń na Ceny Kryptowalut

testWyloguj

Kryptowaluta

Od

Do

Kategoria

BTC

▼

01.01.2020

22.09.2022

Regulacje/Prawo

▼

Zastosuj

Wykres ceny z zaznaczonymi wydarzeniami

BTC USD

Średni zwrot: Dni z wydarzeniami vs kontrolne

Istotność statystyczna (p-wartość, Welch t-test)

Tabela wydarzeń

96 wierszy

Data	Kategoria	Tytuł	Cena (USD)	Zmiana %
2022-05-14	Regulacje/Prawo	13 Shot, 10 Dead In 'Racially Motivated Hate Crime' At Buffalo Supermarket: Sheriff	\$30 101,266	+2.79%
2022-05-08	Regulacje/Prawo	Brooklyn Subway Shooting Suspect Indicted On Terrorism Charge	\$34 059,266	-4.06%
2022-04-16	Regulacje/Prawo	Man, 21, Arrested A Week After 3 Killed At Georgia Gun Range	\$40 424,484	-0.32%
2022-04-16	Regulacje/Prawo	Arrest Made In South Carolina Mall Shooting That Wounded 14 People	\$40 424,484	-0.32%
2022-04-14	Regulacje/Prawo	LA Political Donor Gets 30 Years In Prison For Fetish Deaths	\$39 935,516	-2.99%
2022-04-09	Regulacje/Prawo	Las Vegas Teen Accused Of Attempting To Murder His Teacher	\$42 782,137	+1.17%
2022-03-31	Regulacje/Prawo	DC Police Find 5 Fetuses In Home Of Anti-Abortion Activist Lauren Handy	\$45 538,676	-3.24%
2022-03-20	Regulacje/Prawo	Shooting Wounds 4 Near South By Southwest Festival In Austin	\$41 247,824	-2.23%
2022-03-14	Regulacje/Prawo	Cops In NYC, DC, Suspect Same Gunman Shot 5 Homeless People	\$39 666,754	+4.80%
2022-03-13	Regulacje/Prawo	NYC Police Search For Suspect Who Shot 2 Sleeping Homeless Men, One Fatally	\$37 849,664	-2.71%
2022-02-23	Regulacje/Prawo	2 Louisiana Deputies Charged In Shooting Death Of Man In Parked Car	\$37 296,57	-2.58%
2022-02-22	Regulacje/Prawo	Man Identified In Portland Mass Shooting Railed Against BLM Protesters: Report	\$38 286,027	+3.27%

Import / Eksport

EKSPORT

Eksport JSON (wydarzenia)

Eksport XML (ceny)

IMPORT

Przeglądaj...

Nie wyb... pliku...

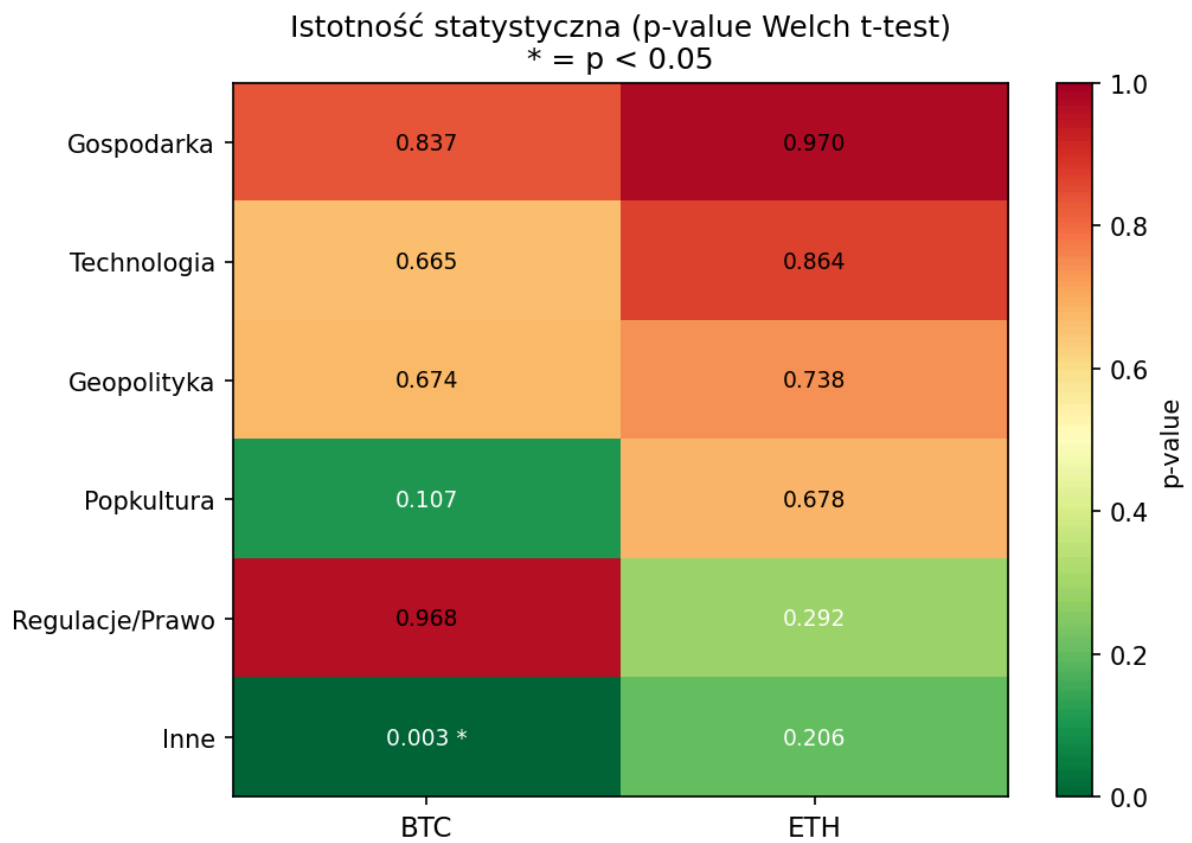
Wyślij XML

Przeglądaj...

Nie wyb... pliku...

Wyślij JSON

Wynik analizy statystycznej



Backend

REST

GET

http://localhost:8000/api/prices?crypto=BTC

Send

Docs

Params

Authorization

Headers 8

Body

Scripts

Tests

Settings

Cookies

raw

JSON

Schema

Beautify

1

Body

200 OK • 117 ms • 179.44 KB • Save Response

JSON

Preview

Visualize

```
1 {
2   "total": 2604,
3   "items": [
4     {
5       "id": 1,
6       "date": "2015-08-07",
7       "price": 279.5849914550781,
8       "symbol": "BTC"
9     },
10    {
11      "id": 2,
12      "date": "2015-08-08",
13      "price": 260.9970092773437,
14      "symbol": "BTC"
15    },
16    {
17      "id": 3,
18      "date": "2015-08-09",
19      "price": 265.0830078125,
20      "symbol": "BTC"
21    },
22    {
23      "id": 4,
24      "date": "2015-08-10"
```

GET http://localhost:8000/api/events?limit=10 Send

Docs Params Authorization Headers 8 Body Scripts Tests Settings Cook

raw JSON Schema Beautify

1

Body 200 OK • 217 ms • 2.19 KB • Save Response

{ } JSON Preview Visualize

```

1  {
2    "total": 10,
3    "limit": 10,
4    "items": [
5      {
6        "id": 1,
7        "title": "Over 4 Million Americans Roll Up Sleeves For Omicron-Targeted
8          COVID Boosters",
9        "description": "Health experts said it is too early to predict whether
10          demand would match up with the 171 million doses of the new boosters
11          the U.S. ordered for the fall.",
12        "date": "2022-09-23",
13        "category": "Inne"
14      },
15      {
16        "id": 2,
17        "title": "American Airlines Flyer Charged, Banned For Life After
18          Punching Flight Attendant On Video",
19        "description": "He was subdued by passengers and crew when he fled to
20          the back of the aircraft after the confrontation, according to the U.
21          S. attorney's office in Los Angeles.",
22        "date": "2022-09-23",
23        "category": "Inne"
24      }
25    ]
26  }

```

SOAP

POST

http://localhost:7000/

Send

Docs

Params

Authorization

Headers 11

Body

Scripts

Tests

Settings

Cooki

raw

JSON

⚠

Schema

Beautify

```
1 <soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
2   xmlns:tns="pl.krypto.soap">
3   <soapenv:Header/>
4   <soapenv:Body>
5     <tns:GetEventCategories/>
6   </soapenv:Body>
7 </soapenv:Envelope>
```

Body

200 OK

85 ms

848 B

Save Response

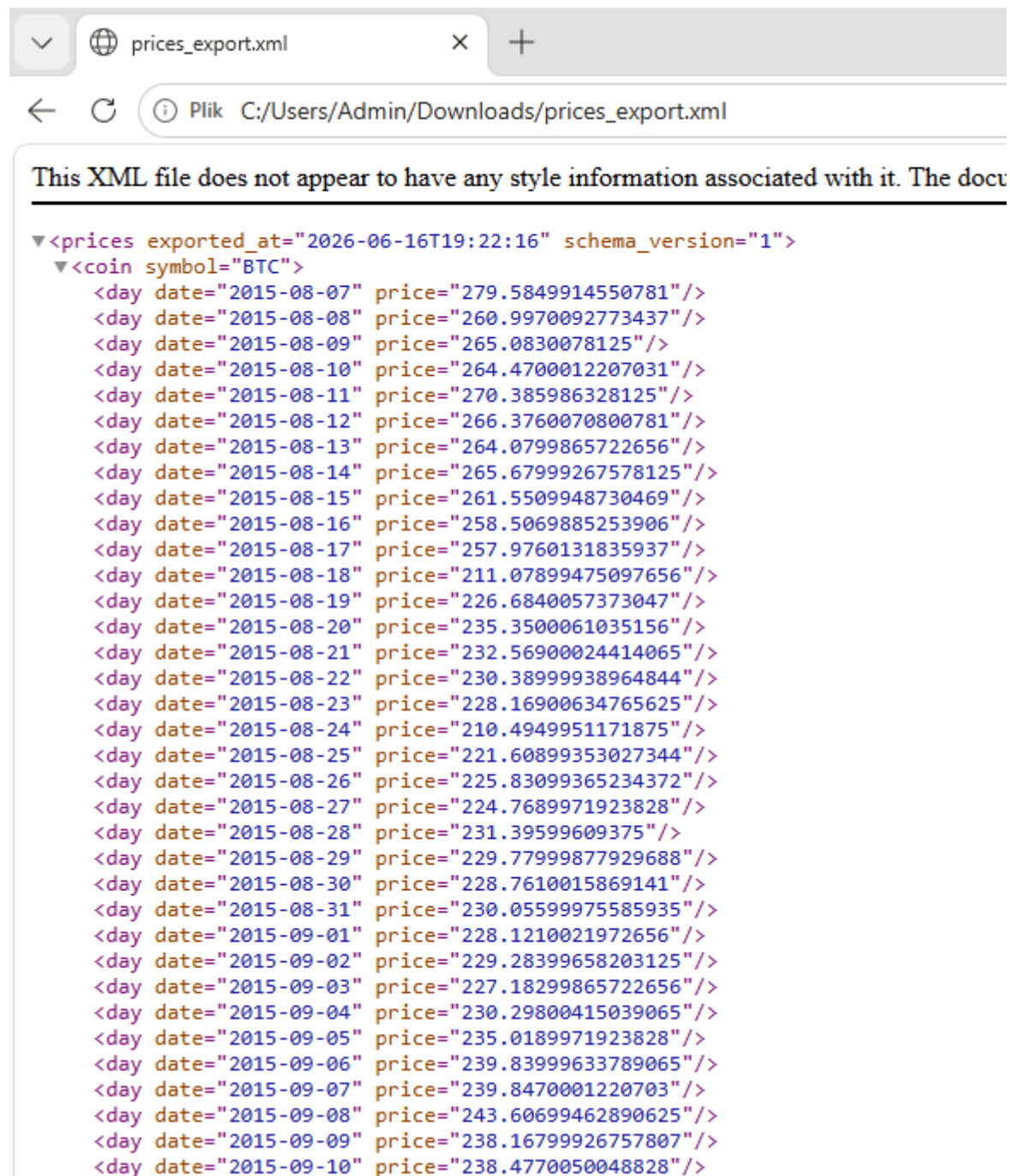
XML

Preview

Visualize

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <soap11env:Envelope xmlns:soap11env="http://schemas.xmlsoap.org/soap/envelope/"
   xmlns:tns="pl.krypto.soap">
3   <soap11env:Body>
4     <tns:GetEventCategoriesResponse>
5       <tns:GetEventCategoriesResult>Geopolityka</tns:GetEventCategoriesResult>
6       <tns:GetEventCategoriesResult>Gospodarka</tns:GetEventCategoriesResult>
7       <tns:GetEventCategoriesResult>Inne</tns:GetEventCategoriesResult>
8       <tns:GetEventCategoriesResult>Popkultura</tns:GetEventCategoriesResult>
9       <tns:GetEventCategoriesResult>Regulacje/Prawo</
       tns:GetEventCategoriesResult>
10      <tns:GetEventCategoriesResult>Technologia</tns:GetEventCategoriesResult>
11    </tns:GetEventCategoriesResponse>
12  </soap11env:Body>
13 </soap11env:Envelope>
```

Przykładowy eksport XML



This XML file does not appear to have any style information associated with it. The document...

```
<prices exported_at="2026-06-16T19:22:16" schema_version="1">
  <coin symbol="BTC">
    <day date="2015-08-07" price="279.5849914550781"/>
    <day date="2015-08-08" price="260.9970092773437"/>
    <day date="2015-08-09" price="265.0830078125"/>
    <day date="2015-08-10" price="264.4700012207031"/>
    <day date="2015-08-11" price="270.385986328125"/>
    <day date="2015-08-12" price="266.3760070800781"/>
    <day date="2015-08-13" price="264.0799865722656"/>
    <day date="2015-08-14" price="265.67999267578125"/>
    <day date="2015-08-15" price="261.5509948730469"/>
    <day date="2015-08-16" price="258.5069885253906"/>
    <day date="2015-08-17" price="257.9760131835937"/>
    <day date="2015-08-18" price="211.07899475097656"/>
    <day date="2015-08-19" price="226.6840057373047"/>
    <day date="2015-08-20" price="235.3500061035156"/>
    <day date="2015-08-21" price="232.56900024414065"/>
    <day date="2015-08-22" price="230.38999938964844"/>
    <day date="2015-08-23" price="228.16900634765625"/>
    <day date="2015-08-24" price="210.4949951171875"/>
    <day date="2015-08-25" price="221.60899353027344"/>
    <day date="2015-08-26" price="225.83099365234372"/>
    <day date="2015-08-27" price="224.7689971923828"/>
    <day date="2015-08-28" price="231.39599609375"/>
    <day date="2015-08-29" price="229.77999877929688"/>
    <day date="2015-08-30" price="228.7610015869141"/>
    <day date="2015-08-31" price="230.05599975585935"/>
    <day date="2015-09-01" price="228.1210021972656"/>
    <day date="2015-09-02" price="229.28399658203125"/>
    <day date="2015-09-03" price="227.18299865722656"/>
    <day date="2015-09-04" price="230.29800415039065"/>
    <day date="2015-09-05" price="235.0189971923828"/>
    <day date="2015-09-06" price="239.83999633789065"/>
    <day date="2015-09-07" price="239.8470001220703"/>
    <day date="2015-09-08" price="243.60699462890625"/>
    <day date="2015-09-09" price="238.16799926757807"/>
    <day date="2015-09-10" price="238.4770050048828"/>
  </coin>
</prices>
```


Przykładowy export JSON

```
{
  "meta": {
    "schema_version": 1,
    "total": 84836
  },
  "records": [
    {
      "date": "2022-09-23",
      "title": "Over 4 Million Americans Roll Up Sleeves For Omicron-Targeted COVID Boosters",
      "category": "Inne"
    },
    {
      "date": "2022-09-23",
      "title": "American Airlines Flyer Charged, Banned For Life After Punching Flight Attendant On Video",
      "category": "Inne"
    },
    {
      "date": "2022-09-23",
      "title": "23 Of The Funniest Tweets About Cats And Dogs This Week (Sept. 17-23)",
      "category": "Popkultura"
    },
    {
      "date": "2022-09-23",
      "title": "The Funniest Tweets From Parents This Week (Sept. 17-23)",
      "category": "Inne"
    },
    {
      "date": "2022-09-22",
      "title": "Woman Who Called Cops On Black Bird-Watcher Loses Lawsuit Against Ex-Employer",
      "category": "Inne"
    },
    {
      "date": "2022-09-22",
      "title": "Cleaner Was Dead In Belk Bathroom For 4 Days Before Body Found: Police",
      "category": "Inne"
    },
    {
      "date": "2022-09-22",
      "title": "Reporter Gets Adorable Surprise From Her Boyfriend While Live On TV",
      "category": "Inne"
    },
    {
      "date": "2022-09-22",
      "title": "Puerto Ricans Desperate For Water After Hurricane Fiona's Rampage",
      "category": "Geopolityka"
    },
    {

```

Zapisy do bazy danych

id	crypto_id	date	price
1	1	2015-08-07	279.5849914550781
2	2	2015-08-08	260.9970092773437
3	3	2015-08-09	265.0830078125
4	4	2015-08-10	264.4700012207031
5	5	2015-08-11	270.385986328125
6	6	2015-08-12	266.3760070800781
7	7	2015-08-13	264.0799865722656
8	8	2015-08-14	265.67999267578125
9	9	2015-08-15	261.5509948730469
10	10	2015-08-16	258.5069885253906
11	11	2015-08-17	257.9760131835937
12	12	2015-08-18	211.07899475097656
13	13	2015-08-19	226.6840057373047
14	14	2015-08-20	235.3500061035156
15	15	2015-08-21	232.56900024414065
16	16	2015-08-22	230.38999938964844

JWT

POST http://localhost:8000/api/auth/login

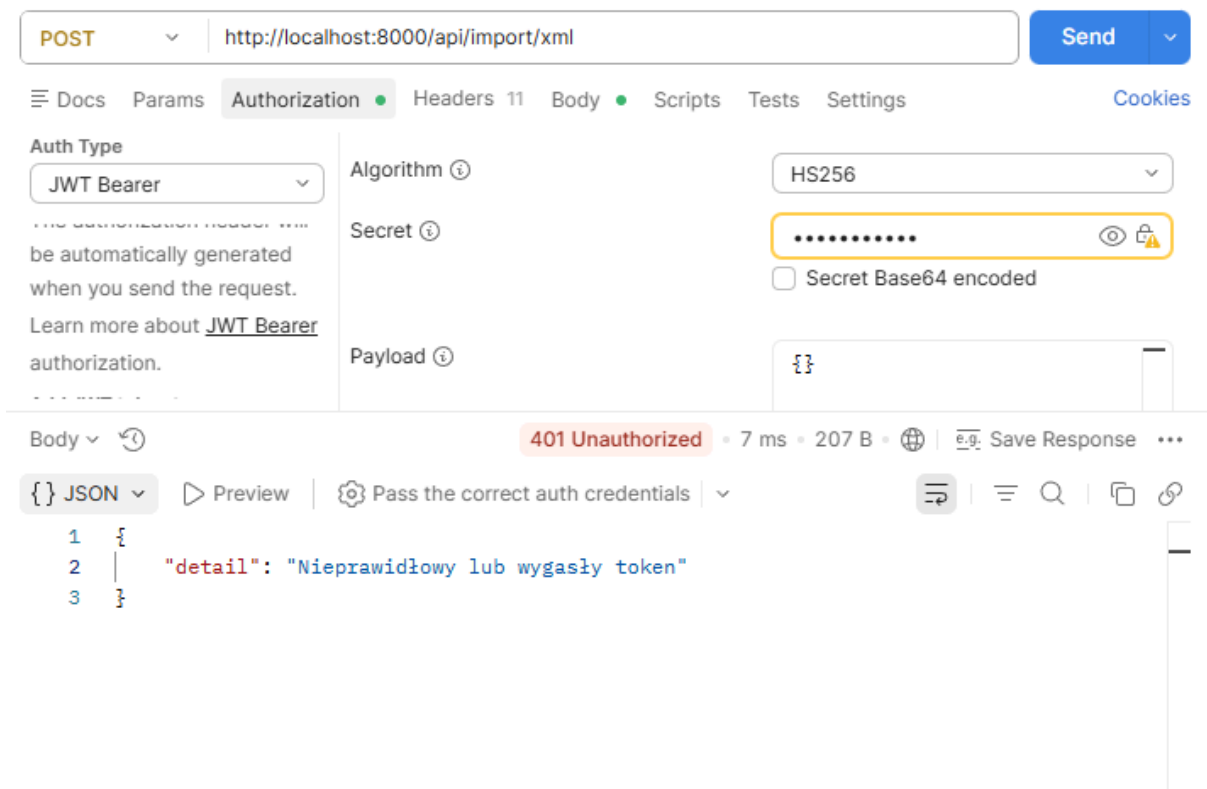
form-data

Key	Value	Description
username	test	
password	admin	

Body

200 OK • 231 ms • 290 B

```
{  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJ0ZXN0IiwiaXhwaWJ0eNzgxNjM0OTAyfQ.mv5RFiEHX07NSrbp19v7naFZ6vbQHm4cgg9Jkzb9IVc",  "token_type": "bearer"}
```



Poziomy izolacji(na oddzielnej tabeli)

```
(.venv) PS C:\Users\Admin\Documents\Integracja systemów\Projekt> python isolation_demo.py

DIRTY READ
PRZED: []
W TRAKCIE: 0
PO: []
Nie ma dirty read w SQLite

LOST UPDATE
PRZED: []
PO: [('BTC', 47000.0, 'T2'), ('BTC', 47500.0, 'T2'), ('BTC', 46800.0, 'T2'), ('BTC', 47000.0, 'T1'), ('BTC', 47500.0, 'T1'), ('BTC', 46800.0, 'T1')]
LICZBA: 6

SERIALIZABLE
PRZED: [('BTC', 47000.0, 'T2'), ('BTC', 47500.0, 'T2'), ('BTC', 46800.0, 'T2'), ('BTC', 47000.0, 'T1'), ('BTC', 47500.0, 'T1'), ('BTC', 46800.0, 'T1')]
WORKERS: ('T2': 'ok', 'T1': 'ok')
PO: [('BTC', 47000.0, 'T2'), ('BTC', 47500.0, 'T2'), ('BTC', 46800.0, 'T2'), ('BTC', 47000.0, 'T1'), ('BTC', 47500.0, 'T1'), ('BTC', 46800.0, 'T1'), ('BTC', 47000.0, 'T2'), ('BTC', 47500.0, 'T2'), ('BTC', 46800.0, 'T2'), ('BTC', 47000.0, 'T1'), ('BTC', 47500.0, 'T1'), ('BTC', 46800.0, 'T1')]
(.venv) PS C:\Users\Admin\Documents\Integracja systemów\Projekt>
```

Mechanizm transakcji w dostępie:

W pliku import_export i api używany jest kontekst Session, rollback jest przeprowadzany automatycznie przy wyjątku jeśli nie było commit() przed db.close().